

OPTIMIZATION OF ZF

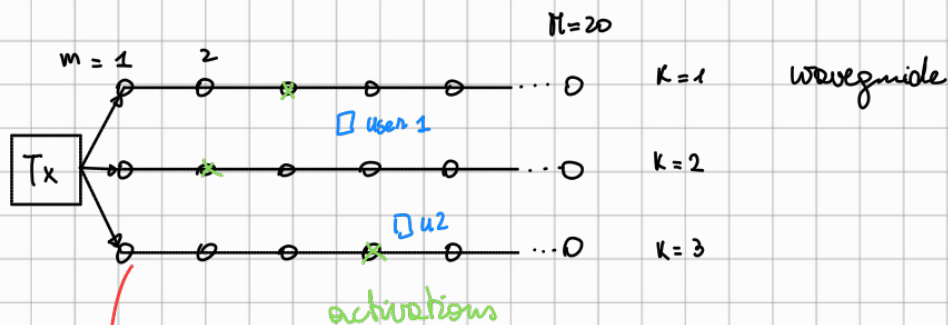
- sum-rate with ZF precoding ($K=3, U=2$)

$$R = \sum_k \log_2 \left(1 + \frac{E_k}{\sigma_n^2} \frac{|\underline{h}_k^H \underline{f}_k|^2}{\|\underline{f}_k\|^2} \right)$$

$$= \sum_k \log_2 \left(1 + \frac{E_k}{\sigma_n^2} \cdot \frac{|\underline{h}_k^H \underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_k|^2}{\|\underline{H}(\underline{H}^H \underline{H})^{-1} \underline{e}_k\|^2} \right)$$

with $\underline{E} = \underline{H}(\underline{H}^H \underline{H})^{-1}$

- scenario



$\underline{S}$ (elements $S_{k,m}$) activation matrix s.t. $\begin{cases} S_{k,m} = \{0, 1\} \\ \sum_{m=1}^M S_{k,m} = 1 \rightarrow 1 \text{ activation per waveguide} \end{cases}$

- problem:

$$\underset{\underline{S}}{\text{maximize}} \{R\} = \underset{\underline{S}}{\text{minimize}} \{-R\}$$

s.t. C1: $S_{k,m} = \{0, 1\}$

C2: $\sum_m S_{k,m} = 1 \quad \forall k$

- relaxation of binary constraint C1 to continuous values

$\overline{C1}: 0 \leq S_{k,m} \leq 1$

$\Rightarrow \underset{\underline{S}}{\text{minimize}} \{-R\} \quad \text{s.t. } \overline{C1}, C2$

JOINT OPTIMIZATION FOR PA POSITION AND BEAMFORMING WEIGHTS

- given number of users U , number of waveguide K and M possible PA positions on every waveguide
- given the channel gain matrix $\underline{C}$ which contains the channel gain coefficients for every antenna position, for every waveguide, for every user:

$$\underline{C} \in \mathbb{C}^{K \times N \times U} \rightarrow \text{elements } C_{k,m,u}$$

$$\text{↳ or } \underline{C} = [\underline{C}_1 \dots \underline{C}_U] \rightarrow \underline{C}_u \in \mathbb{C}^{K \times N}$$

- the channel matrix $\underline{H} \in \mathbb{C}^{K \times U}$ can be obtained through the binary selection matrix $\underline{S} \in \{0,1\}^{K \times N}$. Each channel vector $\underline{h}_u \in \mathbb{C}^{K \times 1}$ for each user is defined as

$$\underline{h}_u = \underline{S} \odot \underline{C}_u \cdot \underline{1}_N$$

and the channel matrix and its elements can be written as

$$\underline{H} = [\underline{h}_1 \dots \underline{h}_U] \quad H[k,u] = \underline{S}[k,:] \cdot \underline{C}[k,:,u]^T$$

- the signal model for user u , with its beamforming vector $\underline{w}_u \in \mathbb{C}^{K \times 1}$, is

$$\begin{aligned} y_u &= \underline{h}_u^H \underline{w}_u x_u + \underline{h}_u^H \sum_{i \neq u} \underline{w}_i x_i + n_u \\ &= (\underline{S} \odot \underline{C}_u \cdot \underline{1}_N) \cdot \sum_{i=1}^U \underline{w}_i x_i + n_u \end{aligned}$$

- SINR for user u and the sum rate are

$$\text{SINR: } \gamma_u = \frac{|\underline{h}_u^H \underline{w}_u|^2}{\sum_{i \neq u} |\underline{h}_u^H \underline{w}_i|^2 + \sigma_n^2} = \frac{|\underline{S} \odot \underline{C}_u \cdot \underline{1}_N|^2}{\sum_{i \neq u} |\underline{S} \odot \underline{C}_i \cdot \underline{1}_N|^2 + \sigma_n^2}$$

$$R = \sum_u \log_2(1 + \gamma_u)$$

- the optimisation problem is defined as:

$$\underset{\underline{S}, \underline{W}}{\text{maximization of } R} = \underset{\underline{S}, \underline{W}}{\text{minimization of } -R} \quad \text{s.t.}$$

$$C1: S_{k,m} \in \{0,1\} \xrightarrow{\text{relaxation}} \overline{C1}: 0 \leq S_{k,m} \leq 1$$

$$C2: \sum_m S_{k,m} = 1 \quad \forall k$$

$$C3: \sum_u \|\underline{w}_u\|^2 \leq P$$

· the problem is solved with alternating optimization

→ fix $\underline{S}$ and optimise on $\underline{W}$

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→ until convergence for R